Supporting Information for "Unraveling the Drivers of Water Shortage across Spatial Scales and Sectors in Colorado's West Slope River Basins"

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Text S1. Historical and Future Demands

The historical demands used in this study are derived from consumptive water demands for each of the users represented in StateMod across all of the West Slope basins. These historical demands exhibit substantial variability both across basins and inter-annual variation, reflecting differences in climate and water management practices (Figure S4). Water users in the West Slope basins are categorized into three sectors: municipal, irrigation, and industrial users. Figure R1 provides additional context by showing the distribution of mean annual sectoral demands aggregated across all West Slope basins. The mean annual historical demand volumes are 53.6 million m^3 for municipal use, 32.3 million m^3 for industrial use, and 2006 million m^3 for irrigation. Clearly, agricultural water use dominates

total demand and is therefore the primary sector driving potential water shortages. A detailed summary of the full range of sampled demands used in our analysis are provided in Figure S4 in our supplemental materials. which are constructed to capture uncertain projections of demands for the mid-century 2050s time period.

Our exploration of future uncertain mid-century demands adopts an Latin Hypercube sampling approach for the lower and upper bound of demand changes listed in Table 1. Specifically, baseline demands of the historical distributions are taken and multiplicative change factors are applied to represent plausible future growth or decline across each of the water users across the West Slope basins. This application of change factors follows the same approach detailed in Hadjimichael et al. (2020). These change factors are designed to capture uncertainty in demand drivers such as population growth, agricultural expansion, industrial growth and other factors. The basin and sector specific ranges sampled in Table 1 are derived from Colorado Water Conservation Board (2023), which provides detailed demand projections under five future scenarios developed in collaboration with diverse stakeholder groups, including the Interbasin Compact Committee, basin roundtables, and the Outreach Working Group. Table 1 synthesizes these projections by extracting the minimum and maximum values across scenarios to define lower and upper bounds, which were then sampled and applied to users across West Slope basins.

Text S2. Quantile sampling selection technique using drought index

As discussed in Section 3.2, for each uncertain factor combination, 1000 streamflow realizations are generated sampling the internal variability. Out of 1000, 20 realizations are selected based on quantile technique of drought index for the analysis. The drought

index used is the severity-based metric defined in Gold et al. (2025). This index is defined using a moving-window threshold method identifying periods when n-year rolling mean log annual flow drops 0.5 times standard deviation below the mean flow of the historical record.

$$drought_t = True \qquad if \qquad \mu_n^t < \mu_{hist} - 0.5\sigma_{hist}; False \qquad otherwise \tag{1}$$

μ_{hist} is the mean flow of the 105-year historical record, and σ_{hist} is the standard deviation of flow in the 105-year historical record. Multiple temporal definitions of drought are explored using equation 1 by varying from 6 years (multi-year drought) to 25 years (multi-decadal drought). For a given drought event, we define the drought severity as the sum of the normalized difference between the annual flow during each year of the drought period and the mean annual flow of the historical record shown an cumulative deficit in equation S2.

$$CumulativeDeficit = \int_{start}^{end} \frac{\mu_{hist} - Y_t}{\sigma_{hist}} dt \tag{2}$$

We compute the drought index for 1,000 realizations corresponding to each uncertain factors sample. The realizations are then sorted in decreasing order, and every 5th quantile is selected, yielding a set of 20 representative samples that capture a broad range of drought conditions across the full ensemble.

Text S3. Deriving Streamflow Variations from Global Climate Model projections

In this study, we represent drier and wetter futures by altering the HMM parameters as described in Section3.2. This adjustment resulted in a mean annual real-space streamflow reduction (across all basins) in the range of -30% to 20% compared to a baseline period from 1980-2010. This ranges reflects the climate change impacts on streamflow in the Upper Colorado River Basin as evaluated in previous studies (Hoerling et al., 2024; Hegewisch et al., 2023). Figure S2(a) presents the projected distribution of mid-century streamflow variations at Lees Ferry based on 220 samples from large ensemble simulations (LENS) conducted by Hoerling et al. (2024), which incorporate climate projections from CMIP6. A precipitation sensitivity of 1.9, estimated from historical observations, was applied along with three plausible temperature sensitivity scenarios (three different histograms in Figure S2(a)) to characterize the range of streamflow responses.

The analysis conducted by Hoerling et al. (2024) employs a baseline period of 2000–2020, which does not exactly overlap with the range used in this study and therefore prevents a direct comparison. Nevertheless, CMIP6 LENS simulations for that period indicate warmer conditions with precipitation patterns similar to those of the 20th century. As a result, the 2000–2020 baseline reflects comparatively optimistic streamflow outcomes relative to the 1980–2010 reference period used in Figure S2(b), since higher temperatures typically correspond to lower streamflows. In this context, the range of 30% reduction to 20% increase in streamflow from the HMM ensemble in this study aligns well with climate change scenarios as represented in the CMIP6 LENS projections.

Text S4. Classifying the hidden states using the Viterbi algorithm

We predict the states (dry or wet) for each year of the historical record for all the West Slope basins using the Viterbi algorithm. The Viterbi algorithm optimizes the probability of observing the true trajectory (Yt,. . . .YT) with a dynamic programming problem using the give alternative possible state trajectories by treating the states at time step t (Xt) as the decision variables. X is the state and Y is the log streamflow. Formally, the algorithm is represented as follows (notation adopted from Quinn et al. (2018) and Gold et al. (2024):

$$V_{1,k} = P(X_1 = k) = P(Y_1 = y_1|X_t = k)\pi_k$$

$$V_{t,k} = P(Y_t = y_t|X_t = k)\pi_k V_{t-1}$$

where the $V_{1,k}$ is the probability that the first state, X_1, was state k and i is the state in each previous time step.

Text S5. Sectoral sensitivity to uncertain factors across basins

While Figure 9 highlights the Southwest basin, similar patterns of sectoral sensitivity to uncertain factors are observed across other basins, with notable differences outlined as follows. (Supplementary Figures S7-S10). Under both drier and wetter future scenarios, the industrial sector in the Gunnison Basin experiences zero shortages (Figure S5 b), likely due to the smaller number of users and the absence of significant projected growth in the region. The municipal sectors in the White and Yampa basins (Figure S6 and S7) also exhibit distinct behavior compared to other basins. For the municipal water users in the White basins, shortages are not observed at certain shortage magnitude percentiles. In the

Yampa basin, this is further complicated by water availability challenges, where shortages are largely driven by hydrological factors such as snowmelt timing and variability.

Table S1. Percentage change in streamflows for the 20000 sampled scenarios relative to the historical baseline (1990-2010) and the number of scenarios under each category of percentage change.

Percent change	Number of scenarios	Percentage of scenarios
$> 20\%$	34	0.17%
15% to 20%	253	1.2%
10% to 15%	932	4.66%
5% to 10%	1777	8.88%
0% to 5%	2548	12.74%
0% to -10%	6649	33.2%
-10% to -20%	6206	31.03%
$< -20\%$	1601	8%

Table S2. F-test as a statistical measure to assess the significance of differences in responses due to individual parameters and their interactions. The p-value for the F-statistic, represents the probability of observing the calculated F-statistic or a more extreme one if the null hypothesis were true.

West Slope basin	**F-statistic**	**P-value**
Upper Colorado	9.4×10^{32}	0.0
Gunnison	6.9×10^{32}	0.0
Yampa	3.4×10^{32}	0.0
White	0.26×10^{32}	0.0
Southwest	30.6×10^{32}	0.0

Table S3. Average historical demands for the selected districts classified as median senior and junior priority rights as shown in Figure 8

Basin	Median senior priority rights	Median junior priority rights
Upper Colorado	116 million m^3	227 million m^3
Gunnison	204 million m^3	16 million m^3
Yampa	24 million m^3	2 million m^3
Southwest	203 million m^3	11 million m^3

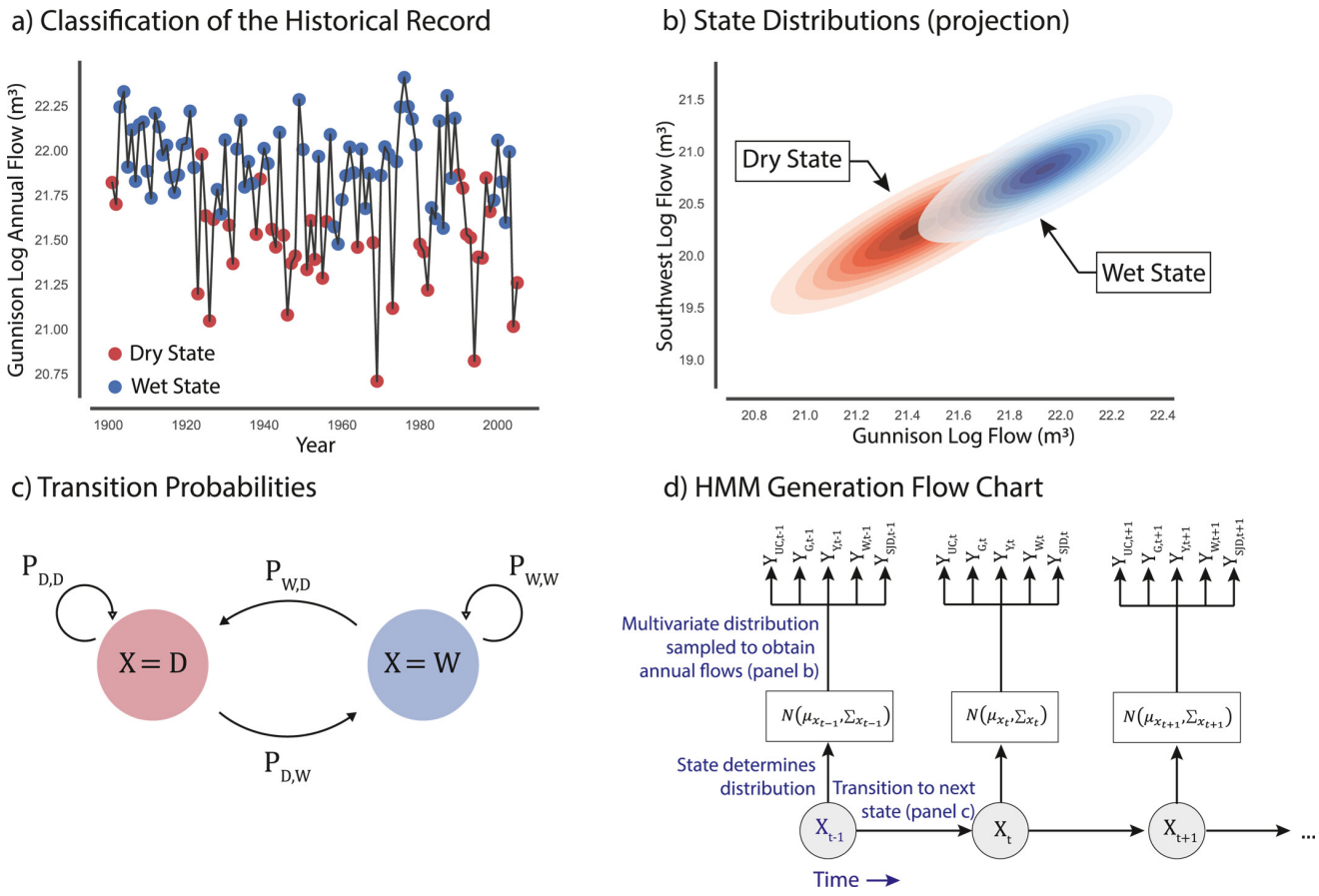


Figure S1. Figure S1. (a) Dry and wet state classification of the historical record in the Gunnison River basin. (b) 2-dimensional projections of the 5-dimensional Gaussian distributions of streamflows of the Gunnison and Southwest Basins based on its dry and wet state. (c) Figure to show transition probabilities from each state. (d) Multi-basin Hidden Markov Model overview used to generate synthetic streamflow records for the West Slope Basins. Figure adopted from Gold et al. (2024).

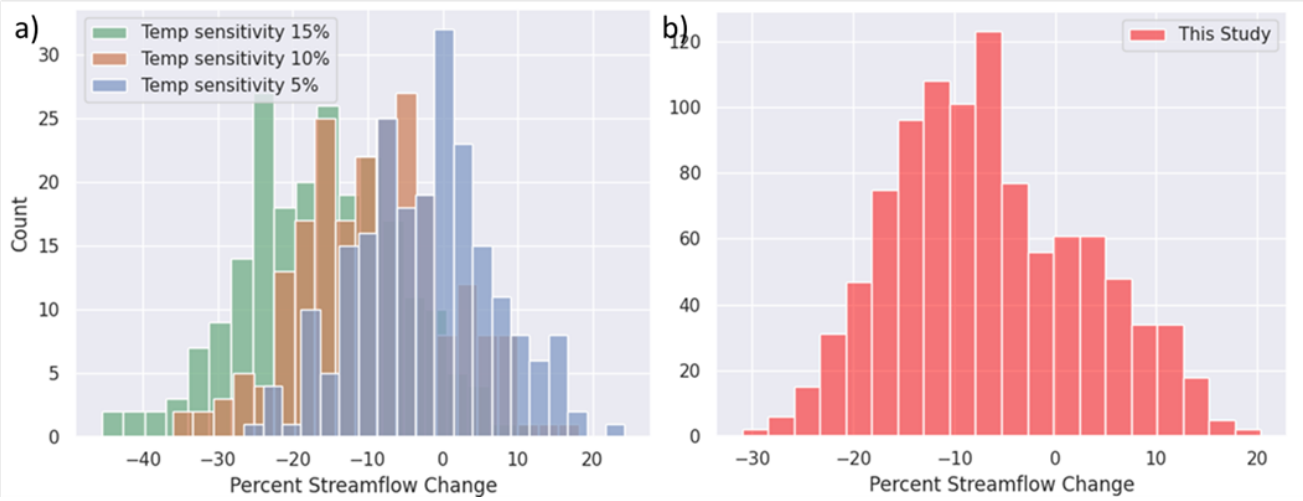


Figure S2. (a) Streamflow changes resulting from large ensemble simulations of mid-century projections evaluated by Hoerling et al. (2024). (b) Streamflow changes sampled in this study to reflect similar ranges from (a) used to evaluate vulnerability measures for the West Slope basins.

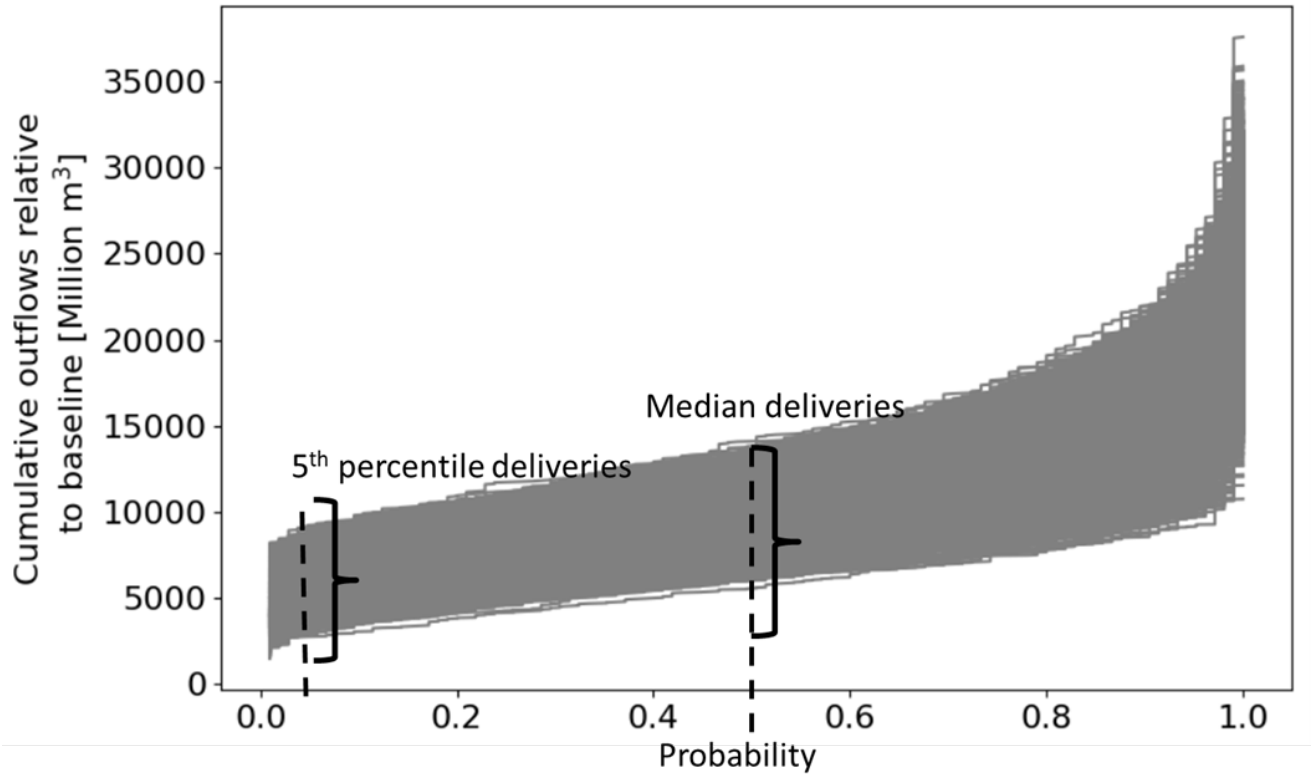


Figure S3. Distribution of cumulative annual outflows (in million m^3) from the West Slope basins under all scenarios relative to historical baseline.

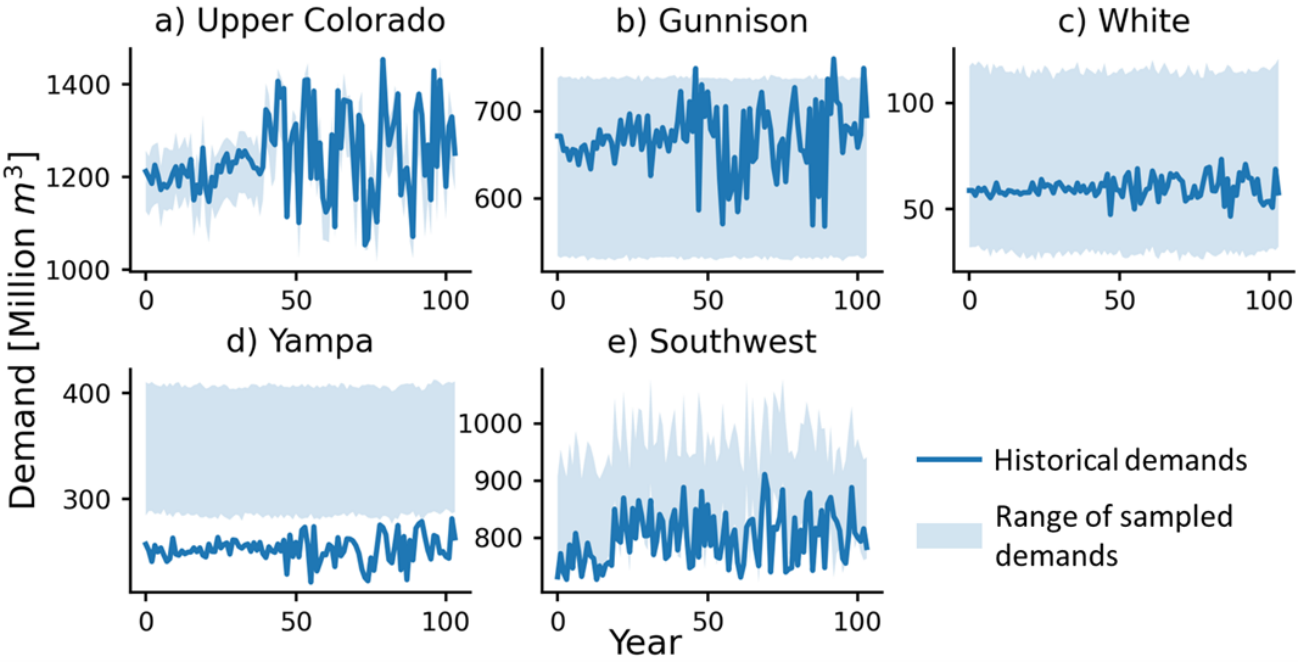


Figure S4. Distribution of historical demands and range of sampled demands across the West Slope basins.

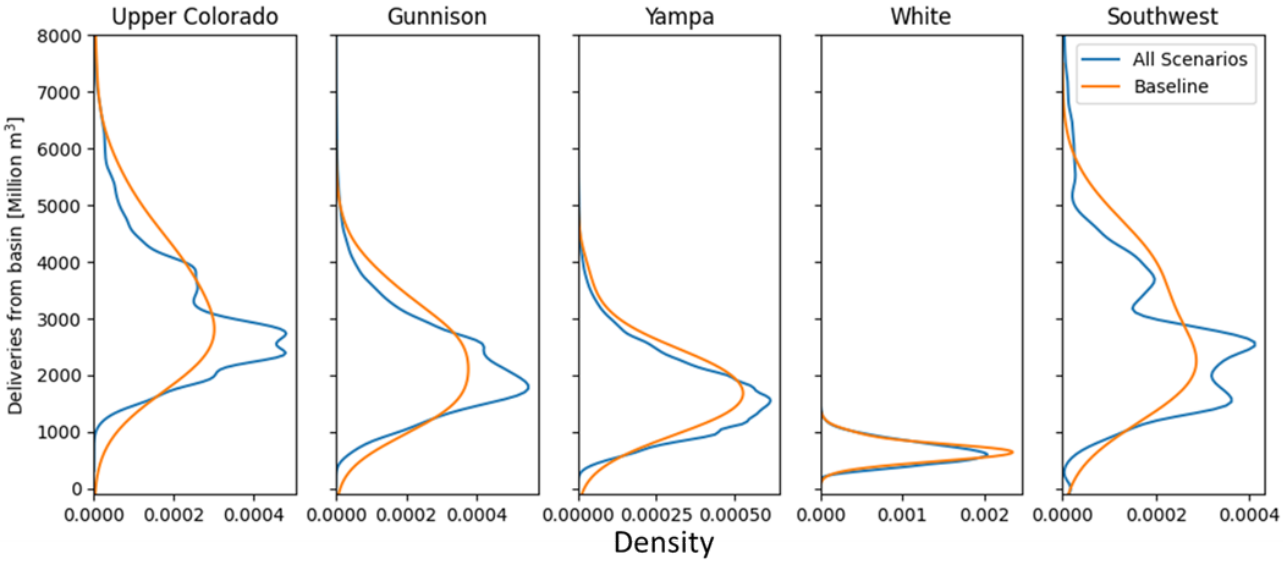


Figure S5. Distributions of deliveries from the West slope basins for historical baseline and all scenarios from the ensemble.

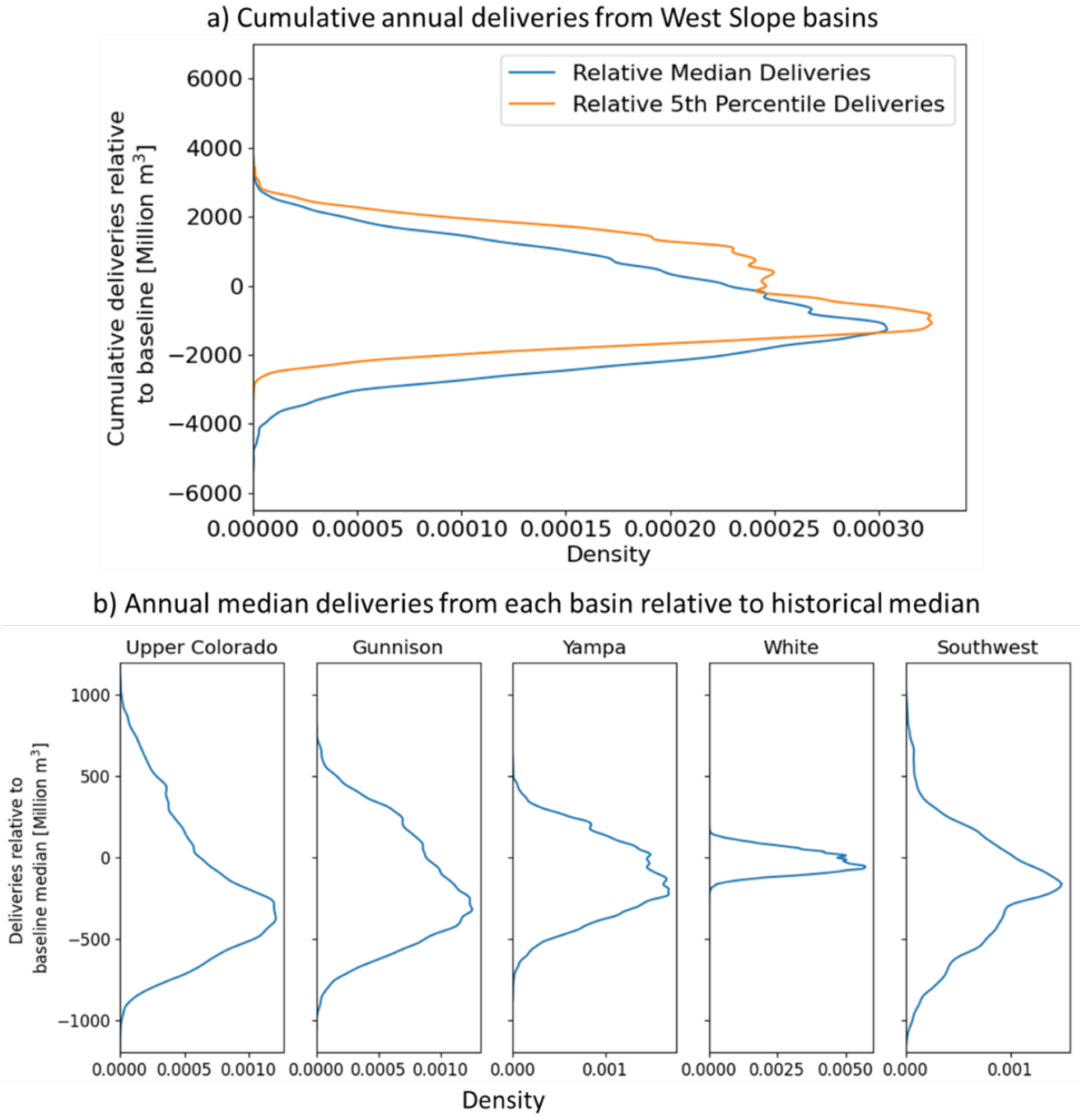


Figure S6. a) Density plots show the distribution baseline deliveries (in million m^3) from the West Slope basins under all scenarios relative to historical baseline. (b) Density plots for difference in median annual deliveries relative to historical median for each of the West Slope basins.

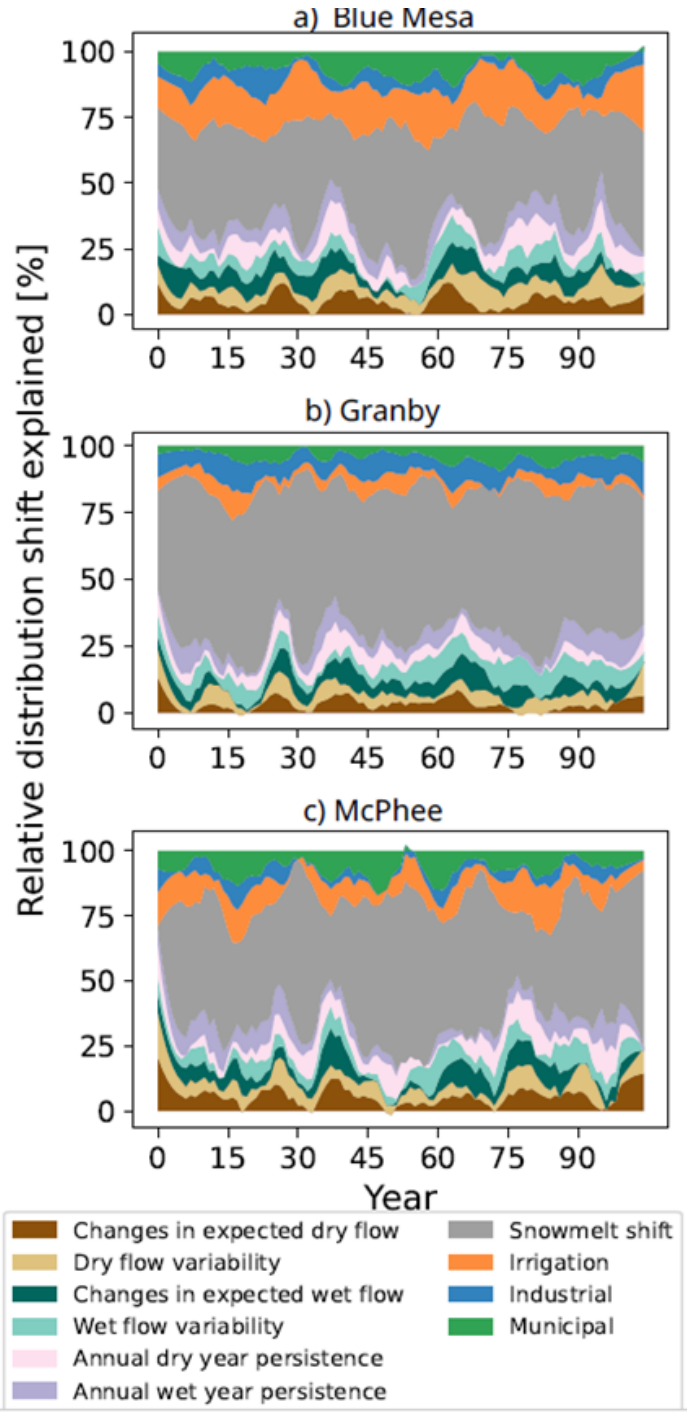


Figure S7. The dominant factors impacting annual storage of the West Slope's three largest reservoirs, are shown in a) Blue Mesa, b) Lake Granby and c) McPhee reservoirs. Each color represents the uncertain factors described in Table 1. The colors in the legend are listed in the order that they are plotted, from bottom, up.

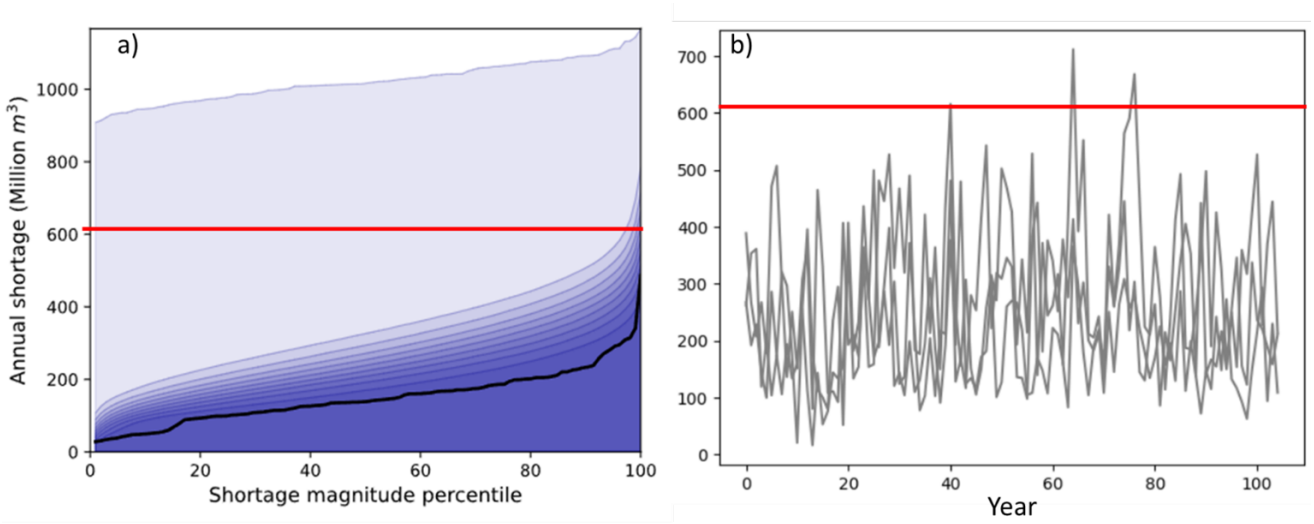


Figure S8. (a) Percentile-varying impacts on cumulative annual shortage magnitude and frequency for the Southwest basin (same as the Figure 6e) (b) Annual shortage magnitude for three randomly selected scenarios in the 105-year timeseries. Redline represents the shortage magnitude of 610 million m3

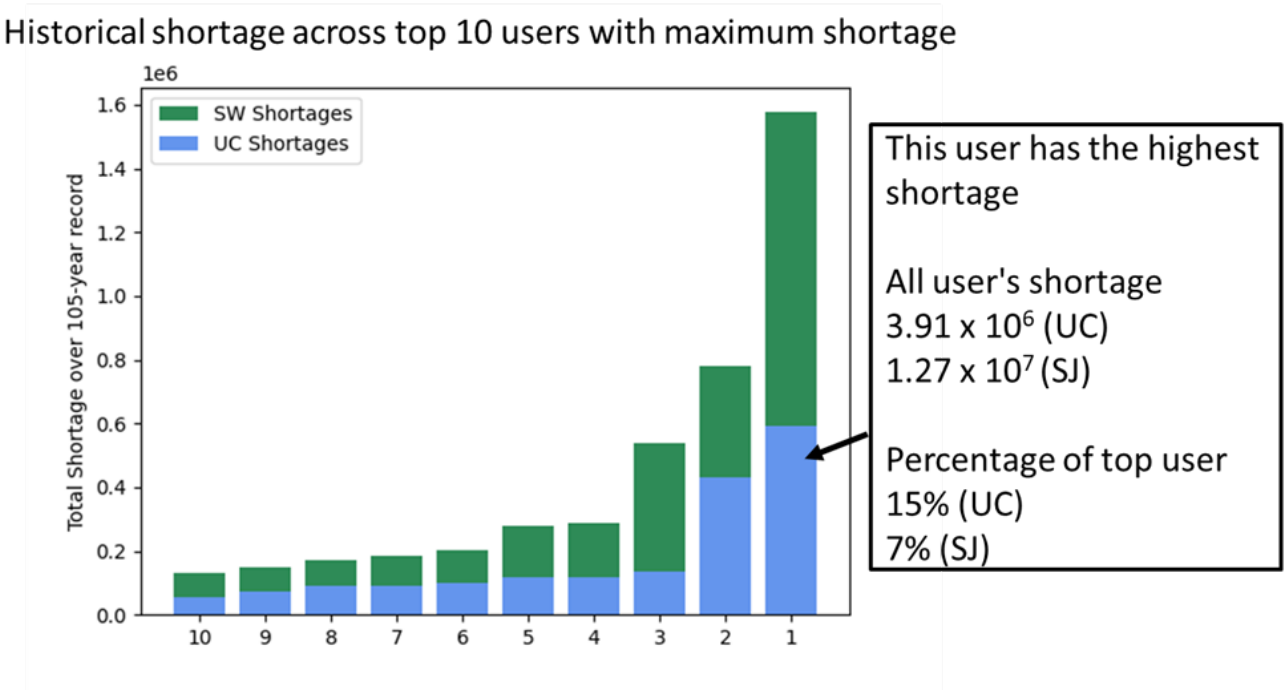


Figure S9. Historical shortage comparisons for top 10 Upper Colorado and Southwest users that contribute the most to total shortage in their respective basins.

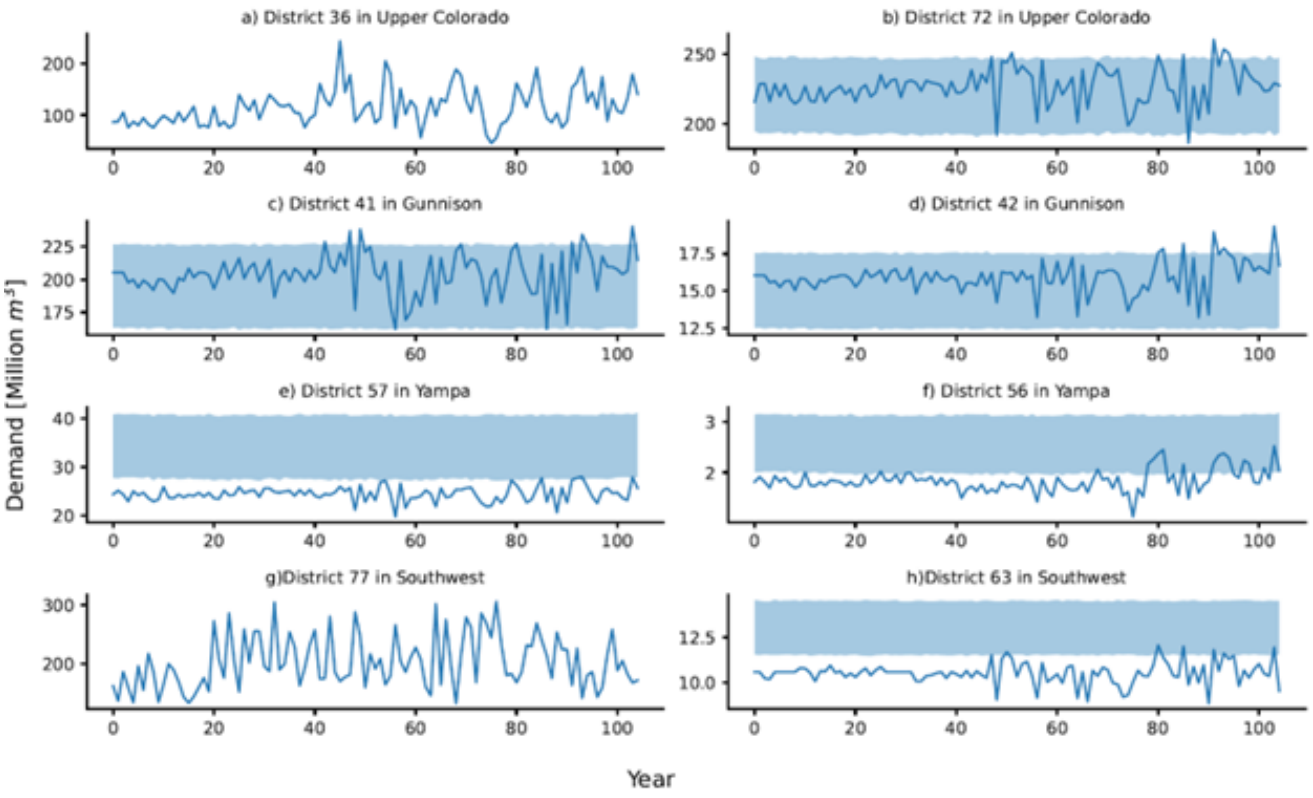


Figure S10. Distribution of historical demands and range of sampled demands across the selected districts of the West Slope basins. The solid blue line is the historical demands, and the shaded regions show the range of sampled demands. For District 36 and District 77, the shaded uncertainty bands are very narrow and closely follow the central series, making them difficult to distinguish at the full figure scale; they become apparent upon zooming into the figure.[PR26.1] Projected demand increases of up to 98% in the Yampa Basin (Table 1) justify the higher volumes future demand scenarios for District 57 and District 56. Similarly, the projected growth in industrial and municipal demands in District 63 reflect the ranges of future demand scenarios sampled for the Southwest basin.

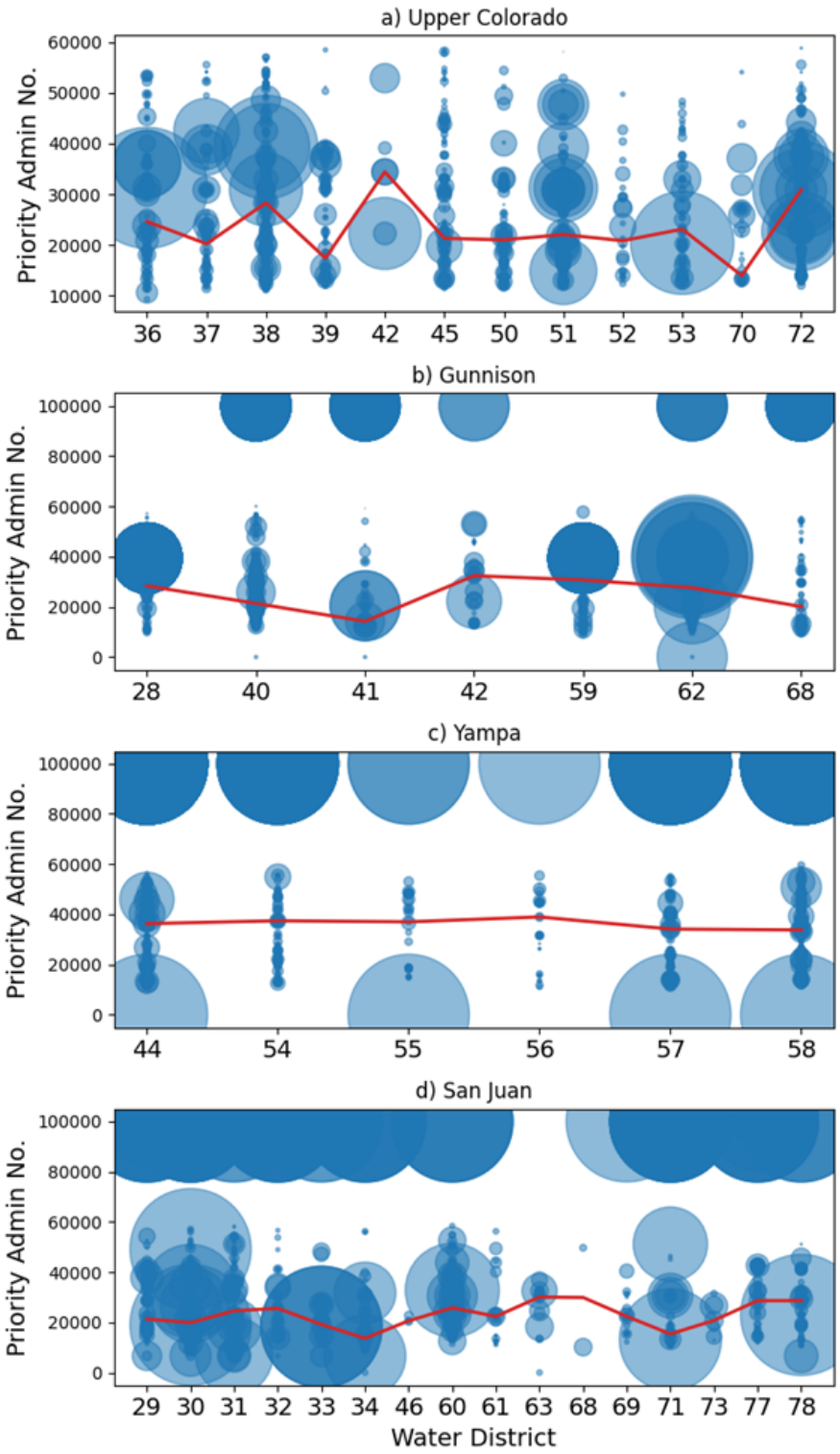


Figure S11. Priority admin number that represents water rights for each users in a water district are shown with bubble size indicating water allocation for the West Slope basins a) Upper Colorado b) Gunnison c) Yampa and d) Southwest. The X-axis represents water districts and y-axis priority admin number. Lower priority admin number is defined for higher right seniority users. Larger bubble size indicates larger decree volumes of users. Red line shows median priority admin numbers across all users within the district.

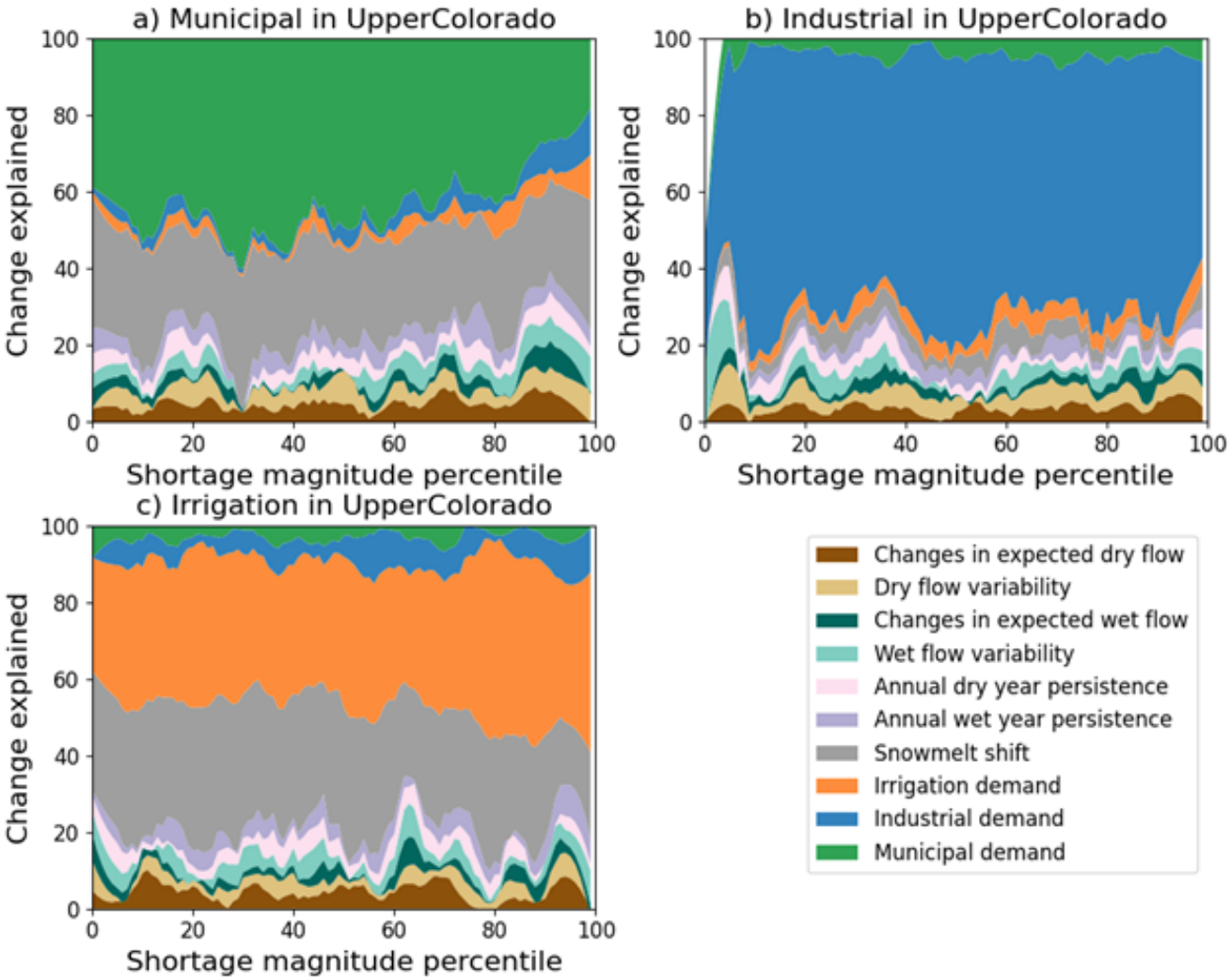


Figure S12. Percentile-varying sensitivity indices of shortage magnitude across different sectors in Upper Colorado basin. X-axis represents the shortage magnitude percentile with increasing in shortage magnitude from 0 to 100. Y-axis represents the percentage of contribution by each uncertain factor. Each panel presents the magnitude-varying sensitivity indices attributed to each factor, for the shortages experienced by (a) Municipal, (b) Industrial and, (c) Irrigation sector. The colors in the legend are listed in the order that they are plotted, from bottom, up.

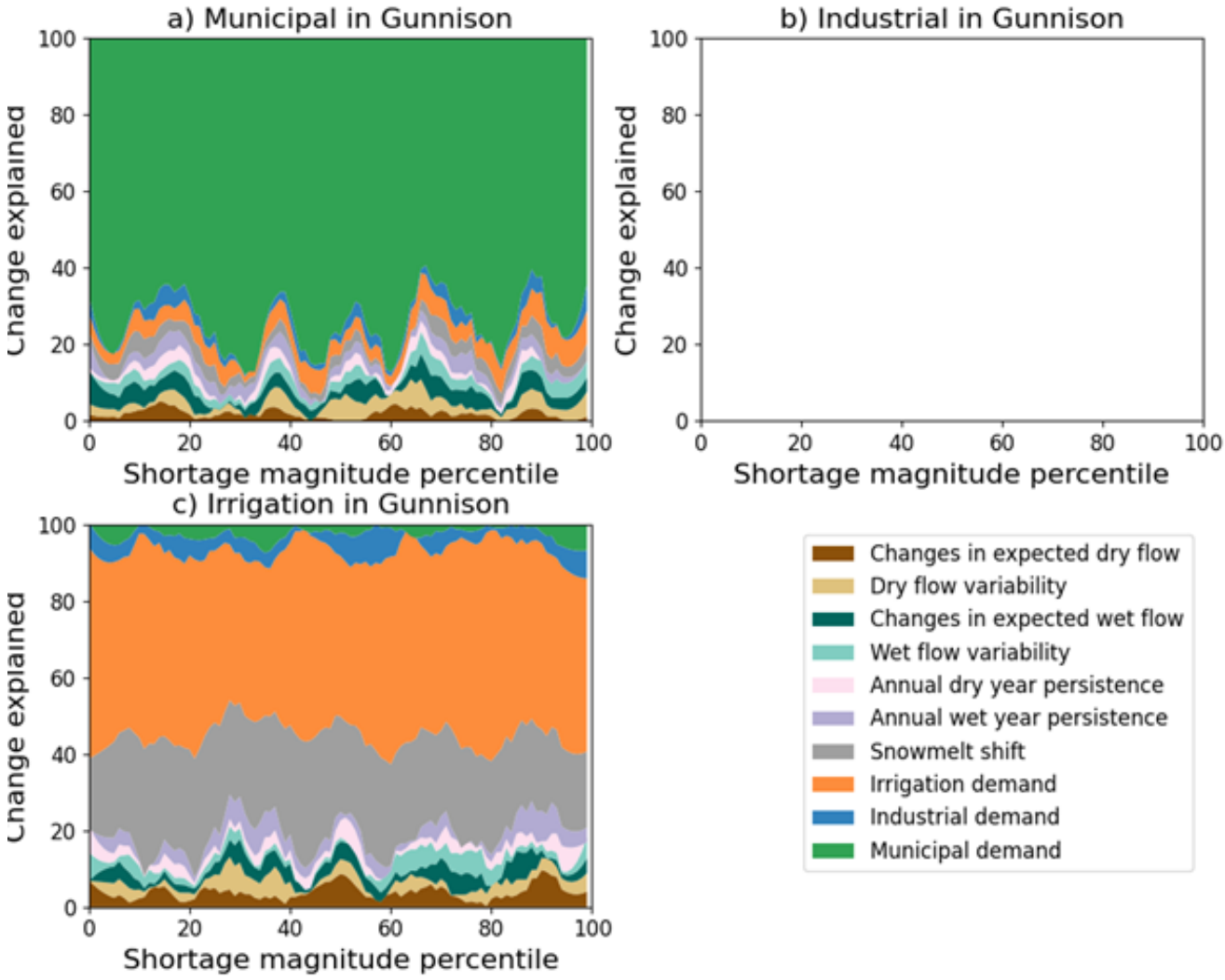


Figure S13. Percentile-varying sensitivity indices of shortage magnitude across different sectors in Gunnison basin. X-axis represents the shortage magnitude percentile with increasing in shortage magnitude from 0 to 100. Y-axis represents the percentage of contribution by each uncertain factor. Each panel presents the magnitude-varying sensitivity indices attributed to each factor, for the shortages experienced by (a) Municipal, (b) Industrial and, (c) Irrigation sector. The colors in the legend are listed in the order that they are plotted, from bottom, up.

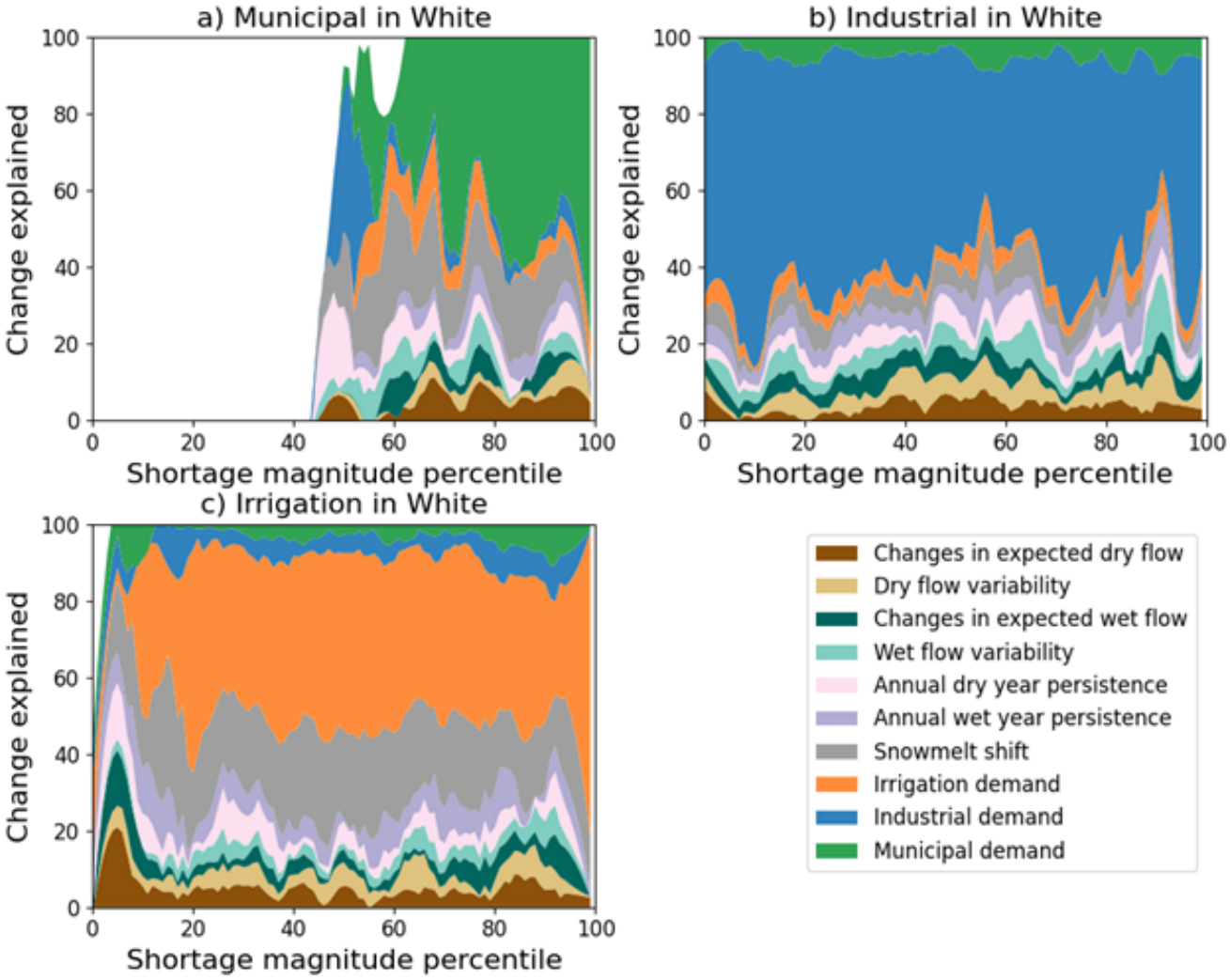


Figure S14. Percentile-varying sensitivity indices of shortage magnitude across different sectors in White basin. X-axis represents the shortage magnitude percentile with increasing in shortage magnitude from 0 to 100. Y-axis represents the percentage of contribution by each uncertain factor. Each panel presents the magnitude-varying sensitivity indices attributed to each factor, for the shortages experienced by (a) Municipal, (b) Industrial and, (c) Irrigation sector. The colors in the legend are listed in the order that they are plotted, from bottom, up.

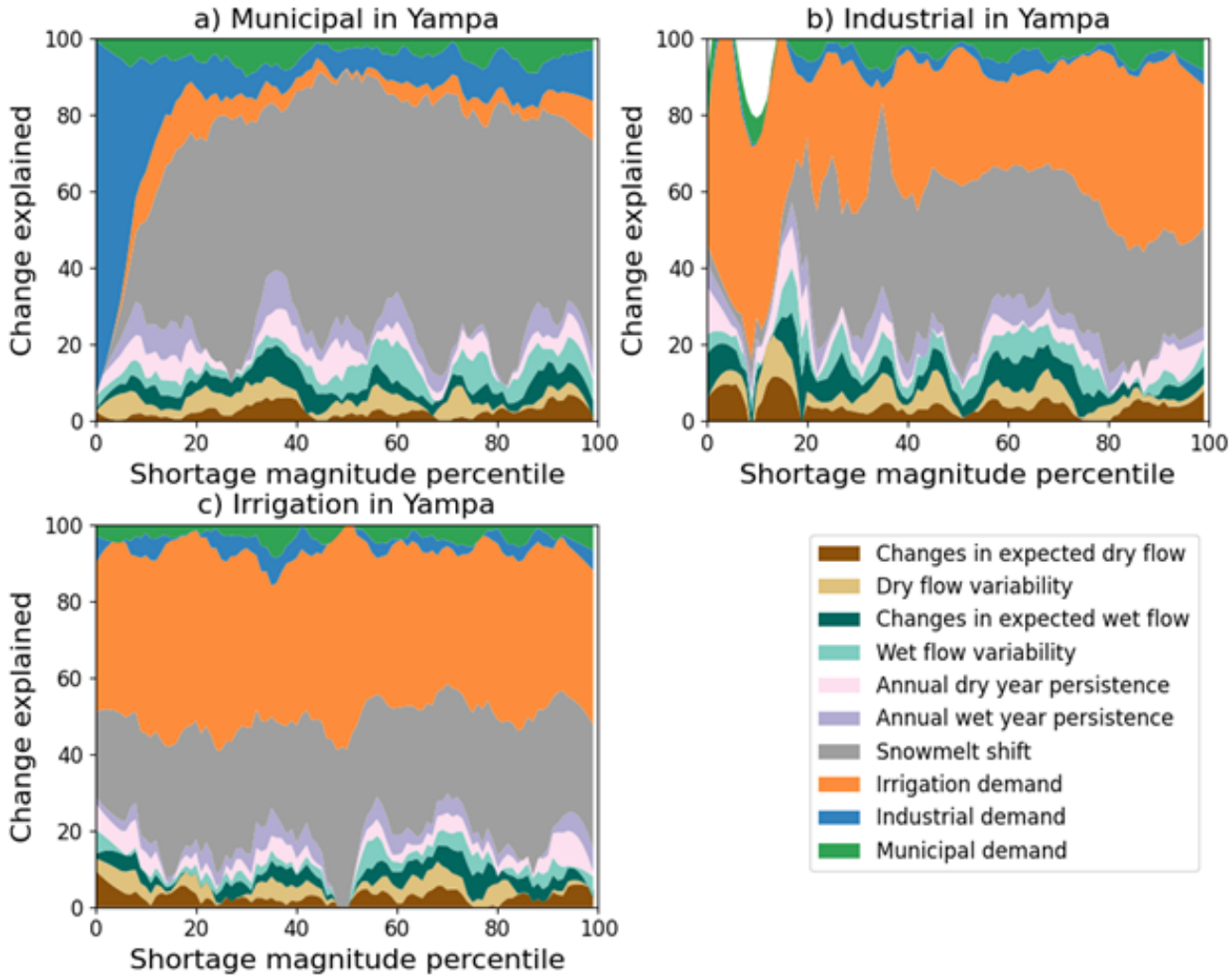


Figure S15. Percentile-varying sensitivity indices of shortage magnitude across different sectors in Yampa basin. X-axis represents the shortage magnitude percentile with increasing in shortage magnitude from 0 to 100. Y-axis represents the percentage of contribution by each uncertain factor. Each panel presents the magnitude-varying sensitivity indices attributed to each factor, for the shortages experienced by (a) Municipal, (b) Industrial and, (c) Irrigation sector. The colors in the legend are listed in the order that they are plotted, from bottom, up.